Superstore Sales Analysis

1. Project Overview

This project analyzes the Superstore dataset to understand business performance across regions, product categories, and customer segments. The objective is to identify key revenue drivers, evaluate profitability, and uncover factors affecting business growth such as discount strategies, customer behavior, and regional performance.

2. Dataset Summary

- **Rows:** 9,994
- **Columns:** 21

Key Features:

- **Order Details:** Order ID, Order Date, Ship Date, Ship Mode
- **Customer Information:** Customer ID, Customer Name, Segment
- **Geographical Data:** Country, City, State, Region
- **Product Information:** Category, Sub-Category, Product Name
- **Business Metrics:** Sales, Quantity, Discount, Profit

Data Types:

- Numerical: Sales, Profit, Quantity, Discount
- Categorical: Region, Category, Segment, Ship Mode
- Date: Order Date, Ship Date

Data Quality:

- No missing values detected
- Dataset is clean and ready for analysis

3. Exploratory Data Analysis (EDA) using Python

Initial exploration was performed to understand the structure, distribution, and key patterns in the dataset.

Data Loading :

- Loaded the Superstore dataset using pandas for analysis

index	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0.0
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3	0.0
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0.0
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2

Initial Exploration

- Checked dataset structure, data types, and column information

```
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   datetime64[ns]
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: datetime64[ns](1), float64(3), int64(3), object(14)
memory usage: 1.6+ MB
```

Statistical Summary

- Analyzed distribution of key numerical variables (Sales, Profit, Quantity, Discount)
- Observed presence of negative profit values indicating loss-making transactions

Feature Engineering

- Converted date columns into datetime format
- Created Year and Month features for time-based analysis

Data Analysis Using SQL(Business transactions)

1. Revenue by Region (Trend Overview)

Compared total revenue generated across different regions to identify the highest-performing markets.

	region character varying (20)	total_revenue numeric
1	West	725457.93
2	East	678781.36
3	Central	501239.88
4	South	391721.90

Findings : West region generates the highest revenue, while South is the lowest-performing region.

2. Revenue vs Profit by Category

Analyzed total revenue, profit, and profit margins across product categories to evaluate overall category performance and profitability.

	category character varying (50)	total_revenue numeric	total_profit numeric	profit_margin_pct numeric
1	Technology	836154.10	145455.66	17.40
2	Furniture	741999.98	18451.25	2.49
3	Office Supplies	719046.99	122490.88	17.04

Findings : Technology and Office Supplies are highly profitable, whereas Furniture has very low profit margins.

3. Revenue-Profit Gap by Region and Category

Examined revenue, profit, and profit margins across regions and categories to identify underperforming areas with low profitability.

	region character varying (20)	category character varying (50)	total_revenue numeric	total_profit numeric	profit_margin_pct numeric
1	Central	Furniture	163797.26	-2871.10	-1.75
2	East	Furniture	208291.17	3046.20	1.46
3	West	Furniture	252612.87	11504.92	4.55
4	Central	Office Supplies	167026.33	8879.95	5.32
5	South	Furniture	117298.68	6771.23	5.77
6	South	Technology	148771.91	19991.96	13.44
7	South	Office Supplies	125651.31	19986.52	15.91
8	West	Technology	251991.86	44303.98	17.58
9	East	Technology	264974.04	47462.12	17.91
10	Central	Technology	170416.29	33697.60	19.77
11	East	Office Supplies	205516.15	41014.52	19.96
12	West	Office Supplies	220853.20	52609.89	23.82

Findings: Furniture category in certain regions (especially Central) results in losses, while Technology maintains strong margins across regions.

4. Top Customers by Spending and Category

Identified the top 10 highest-spending customers along with their preferred product categories to support targeted marketing strategies.

	customer_name character varying (100)	category character varying (50)	total_spend numeric
1	Sean Miller	Technology	23481.51
2	Tamara Chand	Technology	17997.95
3	Raymond Buch	Technology	14265.42
4	Tom Ashbrook	Technology	13709.96
5	Sanjit Chand	Office Supplies	12081.26
6	Hunter Lopez	Technology	11640.81
7	Adrian Barton	Office Supplies	11489.25
8	Christopher Conant	Technology	11322.32
9	Ken Lonsdale	Office Supplies	9654.95
10	Sanjit Engle	Technology	9330.00

Findings: Top customers are heavily concentrated in the Technology category, indicating high-value purchases are tech-driven.

Customer Analysis

5. Customer Purchase Frequency Analysis

Analyzed average purchase frequency and segmented customers into one-time and repeat buyers to understand customer retention behavior.

	avg_orders_per_customer numeric	one_time_buyers bigint	repeat_buyers bigint
1	6.32	12	781

Findings : Majority of customers are repeat buyers, showing strong retention and recurring purchase behavior.

6.Average Order Value by Segment

Calculated average order value across customer segments to determine which segments contribute higher-value transactions.

	segment character varying (50)	avg_order_value numeric
1	Home Office	472.67
2	Corporate	466.41
3	Consumer	449.11

Findings : Home Office segment has the highest order value, followed by Corporate, then Consumer.

7. Customer Lifetime Value by Segment

Estimated customer lifetime value by segment by analyzing total revenue, profit, and customer count to identify the most valuable customer groups.

	segment character varying (50)	total_revenue numeric	total_profit numeric	total_customers bigint	lifetime_value_per_customer numeric
1	Corporate	706146.44	91979.45	236	2992.15
2	Home Office	429653.29	60299.01	148	2903.06
3	Consumer	1161401.34	134119.33	409	2839.61

Findings : Corporate customers generate the highest lifetime value, making them the most valuable segment.

8. Best-Selling Products and Profitability

Identified the top 10 best-selling products by quantity and evaluated their revenue and profitability contribution.

	sub_category character varying (50)	total_sales numeric	total_profit numeric	profit_margin_pct numeric
1	Phones	330007.10	44516.25	13.49
2	Chairs	328449.13	26590.15	8.10
3	Storage	223843.59	21279.05	9.51
4	Tables	206965.68	-17725.59	-8.56
5	Binders	203412.77	30221.64	14.86
6	Machines	189238.68	3384.73	1.79
7	Accessories	167380.31	41936.78	25.05
8	Copiers	149528.01	55617.90	37.20
9	Bookcases	114880.05	-3472.56	-3.02
10	Appliances	107532.14	18138.07	16.87
11	Furnishings	91705.12	13059.25	14.24
12	Paper	78479.24	34053.34	43.39
13	Supplies	46673.52	-1188.99	-2.55
14	Art	27118.80	6527.96	24.07
15	Envelopes	16476.38	6964.10	42.27
16	Labels	12486.30	5546.18	44.42
17	Fasteners	3024.25	949.53	31.40

Findings : High-selling products do not always generate high profit, indicating volume ≠ profitability.

Product Analysis

9. High Sales but Low Profit Sub-Categories

	product_name character varying (255)	total_quantity bigint	total_revenue numeric	total_profit numeric
1	Staples	215	755.47	292.24
2	Staple envelope	170	1686.79	751.60
3	Easy-staple paper	150	2504.18	1096.00
4	Staples in misc. colors	86	478.82	124.06
5	KI Adjustable-Height Table	74	4552.62	-248.50
6	Avery Non-Stick Binders	71	217.31	43.56
7	Storex Dura Pro Binders	71	278.57	50.55
8	GBC Premium Transparent Covers with Diagonal Lined Patt...	67	780.45	49.51
9	Situations Contoured Folding Chairs, 4/Set	64	3641.26	234.24
10	Staple-based wall hangings	62	422.29	121.98

Analyzed sub-categories with high sales but low profit margins to identify potential loss-leading products.

Findings : Some sub-categories generate strong sales but have very low or negative profit margins.

10. Discount Impact on Product Profitability

Evaluated products with high average discounts to identify those negatively impacting overall profitability.

	product_name character varying (255)	avg_discount numeric	total_sales numeric	total_profit numeric
1	Cubify CubeX 3D Printer Double Head Print	0.53	11099.96	-8879.97
2	Lexmark MX611dhe Monochrome Laser Printer	0.40	16829.90	-4589.97
3	Cubify CubeX 3D Printer Triple Head Print	0.50	7999.98	-3839.99
4	Bush Advantage Collection Racetrack Conference Table	0.35	9544.72	-1934.40
5	GBC DocuBind P400 Electric Binding System	0.45	17965.07	-1878.17
6	Cisco TelePresence System EX90 Videoconferencing Unit	0.50	22638.48	-1811.08
7	BoxOffice By Design Rectangular and Half-Moon Meeting Room Ta...	0.48	1706.26	-1148.44
8	Epson TM-T88V Direct Thermal Printer - Monochrome - Desktop	0.55	1212.71	-1057.23
9	O'Sullivan 4-Shelf Bookcase in Odessa Pine	0.42	2740.20	-975.12
10	Zebra GK420t Direct Thermal/Thermal Transfer Printer	0.70	703.71	-938.28

Findings : Products with high discounts tend to generate losses, showing discounts negatively impact profit.

11. Profitability per Unit by Category

Calculated profit per unit sold across categories to determine the most efficient and profitable product segments.

	category character varying (50)	total_units bigint	total_profit numeric	profit_per_unit numeric
1	Technology	6939	145455.66	20.96
2	Office Supplies	22906	122490.88	5.35
3	Furniture	8028	18451.25	2.30

Findings: Technology has the highest profit per unit, while Furniture has the lowest efficiency.

12. Month-over-Month Revenue Growth

Analyzed monthly revenue trends and calculated growth rates to identify business expansion patterns and seasonality.

	month text	total_revenue numeric	prev_month_revenue numeric	growth_rate_pct numeric
1	2014-01	14236.90	[null]	[null]
2	2014-02	4519.92	14236.90	-68.25
3	2014-03	55691.04	4519.92	1132.12
4	2014-04	28295.35	55691.04	-49.19
5	2014-05	23648.28	28295.35	-16.42
6	2014-06	34595.14	23648.28	46.29
7	2014-07	33946.37	34595.14	-1.88
8	2014-08	27909.47	33946.37	-17.78
9	2014-09	81777.34	27909.47	193.01
10	2014-10	31453.37	81777.34	-61.54
11	2014-11	78628.74	31453.37	149.99
12	2014-12	69545.64	78628.74	-11.55
13	2015-01	18174.08	69545.64	-73.87
14	2015-02	11951.40	18174.08	-34.24
15	2015-03	38726.26	11951.40	224.03
16	2015-04	34195.25	38726.26	-11.70
17	2015-05	30131.72	34195.25	-11.88
18	2015-06	24797.31	30131.72	-17.70
19	2015-07	28765.32	24797.31	16.00
20	2015-08	36898.32	28765.32	28.27
21	2015-09	64595.87	36898.32	75.06

Findings : Revenue fluctuates significantly across months, indicating inconsistent growth patterns. No consistent seasonal growth pattern is visible, indicating demand is irregular.

Growth Metrics

13. Yearly Revenue Growth by Region

Compared yearly revenue across regions to identify the fastest-growing markets over time.

	region character varying (20)	year numeric	total_revenue numeric
1	Central	2014	103838.16
2	Central	2015	102874.24
3	Central	2016	147429.42
4	Central	2017	147098.06
5	East	2014	128680.45
6	East	2015	156332.01
7	East	2016	180685.95
8	East	2017	213082.95
9	South	2014	103845.88
10	South	2015	71359.97
11	South	2016	93610.23
12	South	2017	122905.82
13	West	2014	147883.07
14	West	2015	139966.24
15	West	2016	187480.26
16	West	2017	250128.36

Findings : West and East show consistent revenue growth, while other regions grow slower or inconsistently.

14. Year-over-Year Profit Margin Trend

Analyzed changes in profit margins over time to evaluate long-term business profitability.

	year numeric	total_revenue numeric	total_profit numeric	profit_margin_pct numeric
1	2014	484247.56	49544.06	10.23
2	2015	470532.46	61618.69	13.10
3	2016	609205.86	81795.27	13.43
4	2017	733215.19	93439.77	12.74

Findings : Profit margins improved over time but dropped in the most recent year.

15. Revenue Growth vs Profit Margin Decline by Category

Identified categories where revenue is increasing but profit margins are shrinking, indicating potential inefficiencies.

	category character varying (50)	year numeric	total_revenue numeric	profit_margin_pct numeric
1	Furniture	2014	157192.89	3.47
2	Furniture	2015	170518.26	1.77
3	Furniture	2016	198901.55	3.50
4	Furniture	2017	215387.28	1.40
5	Office Supplies	2014	151776.41	14.89
6	Office Supplies	2015	137233.42	18.29
7	Office Supplies	2016	183940.07	19.06
8	Office Supplies	2017	246097.09	16.15
9	Technology	2014	175278.26	12.26
10	Technology	2015	162780.78	20.58
11	Technology	2016	226364.24	17.57
12	Technology	2017	271730.82	18.65

Findings: Some categories show increasing revenue but declining profit margins, indicating inefficiencies.

16. Discount vs Profit Relationship

Analyzed how different discount levels impact sales and profitability to understand the effectiveness of discount strategies.

	discount_bucket text	total_orders bigint	avg_sales numeric	total_profit numeric	avg_profit numeric
1	High (40%+)	933	137.87	-99558.86	-106.71
2	Low (0-20%)	3803	222.59	100786.35	26.50
3	Medium (20-40%)	460	509.00	-35817.58	-77.86
4	No Discount	4798	226.74	320987.88	66.90

Findings : Higher discount levels lead to lower profits and even losses, while no-discount sales are most profitable.

Cross-Cutting

17. States with High Orders but Low Profitability

Identified states with high order volumes but poor profit margins to highlight regions with operational inefficiencies.

	state character varying (50)	total_orders bigint	total_revenue numeric	total_profit numeric	profit_margin_pct numeric
1	California	1021	457687.68	76381.60	16.69
2	New York	562	310876.20	74038.64	23.82
3	Texas	487	170187.98	-25729.29	-15.12
4	Pennsylvania	288	116512.02	-15560.04	-13.35
5	Illinois	276	80166.16	-12607.89	-15.73
6	Washington	256	138641.29	33402.70	24.09
7	Ohio	236	78258.21	-16971.37	-21.69
8	Florida	200	89473.73	-3399.25	-3.80
9	North Carolina	136	55603.09	-7490.81	-13.47
10	Michigan	117	76269.61	24463.15	32.07

Findings : Some states generate high orders but low or negative profit, impacting overall performance.

18. Shipping Time Analysis by Mode

Evaluated average shipping time across different shipping modes to assess delivery efficiency and performance.

	ship_mode character varying (50)	avg_shipping_days numeric	total_orders bigint
1	Same Day	0.04	264
2	First Class	2.18	787
3	Second Class	3.24	964
4	Standard Class	5.01	2994

Findings: Standard shipping takes the longest time, while same-day delivery is the fastest.

Summary & Key Business Recommendations

This analysis of 9,994 Superstore transactions reveals critical insights across revenue, customers, products, and growth.

Top 5 Findings:

1. West region leads revenue at $725,457 but South lags significantly at $391,721 — a 46% gap.
2. Technology is the most profitable category at 17.4% margin, while Furniture barely breaks even at 2.49%.
3. Corporate customers have the highest lifetime value at $2,992 per customer despite having fewer customers than the Consumer segment.
4. High discounts (40%+) result in total losses of -$99,558 showing the current discount strategy is damaging profitability.
5. Texas, Ohio, and Illinois generate high order volumes but negative profit margins — indicating operational inefficiencies in these states.

Recommendations:

1. DISCOUNT STRATEGY — Cap discounts at 20%. Orders with no discount generate $320,987 profit vs -$99,558 loss at 40%+ discounts.
2. FURNITURE CATEGORY — Review pricing and supplier costs for Furniture, especially in Central region where it generates negative profit (-$2,871).
3. REGIONAL FOCUS — Invest in South region marketing and sales strategy to close the revenue gap with West and East.
4. CUSTOMER RETENTION — 781 out of 793 customers are repeat buyers. Build a loyalty program to maintain this strong retention rate.
5. STATE-LEVEL ACTION — Conduct operational review for Texas (-15.12% margin), Ohio (-21.69% margin), and Illinois

(-15.73% margin) to identify cost inefficiencies.